

ECEN 743: Reinforcement Learning

Assignment 5: Policy Gradient

Overview

1. You will complete this assignment in a Jupyter notebook. You can download it from our (ECEN743 GitHub repo).
2. You have to submit an HTML report, your code and two videos to Canvas.
3. Put all your files (HTML report, code and videos) into a **single compressed folder** named **LastName.FirstName_A4.zip**.
4. Refer to the **README** of the GitHub repository for how to generate the proper HTML report for submission and for the installation instructions.

In this homework, you will train a policy gradient algorithm to land a lunar lander *with continuous actions* on the surface of the moon. We will use the Lunar Lander environment (LunarLander-v3) from Gymnasium. The environment consists of a lander with continuous actions and a continuous state space. A detailed description of the environment can be found [here](#). The problem is solved if the total reward per **episode is 200 or above**. Do not stop training on the first instance your agent gets a reward above 200, your agent must achieve a reward of 200 or above consistently.

1 Review

Recall the policy gradient theorem(s) from the lecture:

$$\nabla V_{\pi_{\theta}} = \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\cdot)} \left[R(\tau) \sum_{t=0}^{\infty} \nabla \log \pi_{\theta}(s_t, a_t) \right] \quad (1)$$

$$\nabla V_{\pi_{\theta}} = \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t Q_{\pi_{\theta}}(s_t, a_t) \nabla \log \pi_{\theta}(s_t, a_t) \right] \quad (2)$$

$$\nabla V_{\pi_{\theta}} = \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi_{\theta}}(s_t, a_t) \nabla \log \pi_{\theta}(s_t, a_t) \right] \quad (3)$$

We have discussed the motivation behind each of these forms in details during the lectures. Your goal is to code-up each of this form, by replacing the expectation by empirical average appropriately. For example, (1) can be approximated as

$$\nabla V_{\pi_{\theta}} \approx \frac{1}{N} \sum_{i=1}^N \left(R(\tau_i) \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(s_t^i, a_t^i) \right).$$

Similarly, you can approximate (2) as

$$\nabla V_{\pi_{\theta}} \approx \frac{1}{N} \sum_{i=1}^N \left(\sum_{t=0}^T \gamma^t G_t \nabla_{\theta} \log \pi_{\theta}(s_t^i, a_t^i) \right),$$

where $G_t = \sum_{j=t}^T \gamma^{j-t} r(s_j^i, a_j^i)$. A more popular practical approximation is to drop γ^t to get

$$\nabla V_{\pi_\theta} \approx \frac{1}{N} \sum_{i=1}^N \left(\sum_{t=0}^T G_t \nabla_\theta \log \pi_\theta(s_t^i, a_t^i) \right).$$

For approximating $A_{\pi_\theta}(s_t, a_t)$, you can use $G_t - V_{\phi, \pi_\theta}(s_t)$ where ϕ is the parameter of the value network you need to train for a given θ . You can train ϕ with the objective that the following relation is satisfied (approximately)

$$V_\phi(s_t) = \mathbb{E}_{\pi_\theta} \left[\sum_{j=t}^T \gamma^{j-t} r(s_j^i, a_j^i) | s_t \right].$$

2 Questions

1. REINFORCE Algorithm

Implement (1). You should include the training curve in the PDF (x-axis is the number of episodes and the y-axis should be episodic undiscounted-cumulative reward). Use a sliding window average to get smooth plots. Include a description of the hyperparameters used. Try to find the optimal hyperparameters that will enable fast convergence to the optimal policy.

2. Policy Gradient Algorithm

Implement (2) and produce similar results as above.

3. Policy Gradient with Baseline

Implement (3) and produce similar results as above. Also, you should submit a video of the smooth landing achieved by your RL algorithm. Video is needed only for this part.

4. Another Environment

Now, learn the optimal policy for another control problem (environment) using policy gradient algorithm. You can select one environment from the *Classical Control* set, *Box2D* set or *Atari Games* set in Gymnasium. In the HTML report, you need to clearly specify the environment and provide a link to the corresponding page in Gymnasium. You need to include the training curve and describe the hyperparameters used. You should also include the video of the performance.